

Series, Quadratics & Optimization

Series & Summations, Quadratic Equations, Optimization Techniques · Mentor-Led Lab

Developed by TalenciaGlobal

Difficulty ★★★★★ Advanced

Problem

Given a value n , an arithmetic series (first term a , common difference d , k terms) and the coefficients A , B , C of a quadratic $Ax^2 + Bx + C$, compute a set of closed-form sums and then analyse the quadratic.

What to Compute

- Sum of $1..n$ using the closed form $n(n+1)/2$.
- Sum of squares $1^2 + \dots + n^2$ using $n(n+1)(2n+1)/6$.
- Sum of the arithmetic series using $k/2 * (2a + (k-1)d)$.
- The quadratic discriminant $D = B^2 - 4AC$.
- The real roots (two decimals, larger first) when $D \geq 0$, otherwise report that they are complex.
- The vertex $x = -B/(2A)$ and the optimal value $C - B^2/(4A)$; it is a minimum when $A > 0$ and a maximum when $A < 0$.

Input / Output Format

Input: Seven integers in three groups: n ; then a d k ; then A B C .

Output: Labelled lines for Sum $1..n$, Sum of squares, Arithmetic Sum, Discriminant, Roots, Vertex x , and Optimal value (with minimum or maximum).

Examples

Example 1

Input: $n=10$; $a=2$, $d=3$, $k=5$; $A=1$, $B=-5$, $C=6$

Output:

```
Sum 1..n = 55
Sum of squares = 385
Arithmetic Sum = 40
Discriminant = 1
Roots = 3.00 2.00
Vertex x = 2.50
```

Optimal value = -0.25 (minimum)

Explanation: $1+\dots+10 = 55$; squares sum to 385; the series 2,5,8,11,14 sums to 40; $D = 25 - 24 = 1$; roots $(5 \pm 1)/2 = 3$ and 2; vertex at $x = 2.5$ with minimum value $6 - 6.25 = -0.25$.

Example 2

Input: $n=6$; $a=1$, $d=2$, $k=4$; $A=2$, $B=-8$, $C=6$

Output:

```
Sum 1..n = 21
Sum of squares = 91
Arithmetic Sum = 16
Discriminant = 16
Roots = 3.00 1.00
Vertex x = 2.00
Optimal value = -2.00 (minimum)
```

Explanation: sum 21; squares 91; series 1,3,5,7 sums to 16; $D = 64 - 48 = 16$; roots $(8 \pm 4)/4 = 3$ and 1; vertex $x = 2$ with minimum value -2.

Constraints

- $1 \leq n \leq 10^5$
- Coefficient A is non-zero.
- For the sample inputs the discriminant is a perfect square, so the roots are exact.

Approach

Use the closed-form summations rather than loops: $n(n+1)/2$ and $n(n+1)(2n+1)/6$ are exact integer formulas (the division always comes out whole).

The arithmetic series sum $k/2 * (2a + (k-1)d)$ is exact too, because the product $k(2a + (k-1)d)$ is always even.

Compute the discriminant in integers. If it is negative the roots are complex; otherwise take its square root as a double and apply the quadratic formula $(-B \pm \sqrt{D}) / (2A)$.

Optimization: a parabola's extreme point is its vertex at $x = -B / (2A)$. Substituting gives the optimal value $C - B^2 / (4A)$; it is a minimum if the parabola opens upward ($A > 0$) and a maximum if it opens downward ($A < 0$).

Perform the root, vertex and optimal-value divisions in floating point and format the results to two decimals.

Sample Run

Sample Input

```
10
2 3 5
```

```
1 -5 6
```

Sample Output

```
Sum 1..n = 55  
Sum of squares = 385  
Arithmetic Sum = 40  
Discriminant = 1  
Roots = 3.00 2.00  
Vertex x = 2.50  
Optimal value = -0.25 (minimum)
```